

Prompt Engineering

Makeup Examination
Fall 2025–2026

Instructor: Mehmet Ali Akyol, PhD
Ankara Medipol University

Student Name: _____ Student ID: _____

Instructions:

- This exam consists of 20 multiple-choice questions.
 - Each question is worth 5 points for a total of 100 points.
 - Select the **best** answer for each question.
 - You have 60 minutes to complete this exam.
 - No electronic devices or notes are permitted.
 - Mark your answers on the optical form provided.
-

1. A **prompt** in the context of AI is best defined as:
 - A) The internal memory of an LLM
 - B) The training data used to build the model
 - C) The hardware that runs the AI system
 - D) The input message you give to an AI system to elicit a response
2. What is **tokenization** in the context of LLMs?
 - A) The process of encrypting user data
 - B) Generating authentication tokens for API access
 - C) Converting images into text descriptions
 - D) Breaking text into smaller chunks (tokens) that the model can process
3. What are **embeddings** in LLM architecture?
 - A) Physical components of the computer running the model
 - B) The user interface elements of AI applications
 - C) Error logs generated during model training
 - D) Numerical representations of tokens that capture semantic meaning
4. The 2017 paper “Attention Is All You Need” introduced which key innovation?
 - A) Self-attention mechanism (Transformer architecture)
 - B) Tokenization
 - C) Recurrent neural networks

- D) Convolutional neural networks
5. **Hallucination** in LLMs refers to:
- A) The model refusing to answer questions
 - B) The model asking clarifying questions
 - C) The model generating plausible but incorrect or fabricated information
 - D) The model running out of memory
6. In the anatomy of a good prompt, which element specifies headings, JSON schema, or table columns?
- A) Task
 - B) Context
 - C) Constraints
 - D) Output Specification
7. **Zero-shot prompting** is best described as:
- A) Giving just the task without any examples
 - B) Providing multiple examples before the task
 - C) Using images along with text
 - D) Chaining multiple prompts together
8. **Few-shot prompting** is most useful when:
- A) You want to save tokens
 - B) The task is extremely simple
 - C) You need to steer style or format with specific examples
 - D) You want to reduce response time
9. Which is an example of a “prompt anti-pattern”?
- A) Specifying output format clearly
 - B) Providing relevant context
 - C) Setting clear constraints
 - D) Including conflicting directives like “be brief” and “include all details”
10. In Retrieval-Augmented Generation (RAG), what is the primary purpose?
- A) To speed up model inference
 - B) To reduce the model size
 - C) To ground answers with provided excerpts or references
 - D) To train the model on new data
11. The **CO-STAR** framework stands for:
- A) Code, Output, Style, Testing, Analysis, Review
 - B) Context, Objective, Style, Tone, Audience, Response
 - C) Constraint, Optimization, Structure, Target, Action, Result
 - D) Chain, Order, Sequence, Task, Answer, Repeat

12. **Meta-prompting** refers to:
- A) Using prompts to have an LLM generate or refine other prompts
 - B) Prompting about metadata
 - C) Using multiple models simultaneously
 - D) Prompting in multiple languages
13. Which dimension of linguistic register controls terminology density?
- A) Tenor (Relationship)
 - B) Mode (Channel)
 - C) Field (Topic Domain)
 - D) Rhythm
14. **Brand Voice Fingerprinting** involves:
- A) Creating biometric authentication
 - B) Recording audio samples of brand representatives
 - C) Extracting measurable voice attributes from existing content to ensure AI outputs match brand style
 - D) Generating unique identifiers for users
15. What is the main advantage of using “LLM-as-a-Judge” for evaluation?
- A) It is always 100% accurate
 - B) It eliminates the need for any human review
 - C) It enables scalable evaluation of thousands of outputs quickly
 - D) It reduces model training costs
16. **Verbosity Bias** in LLM evaluation refers to:
- A) Models producing too few tokens
 - B) Preference for technical jargon
 - C) The tendency to rate longer answers as better, even if they contain fluff
 - D) Bias toward English-language responses
17. **Tree of Thoughts (ToT)** differs from Chain-of-Thought by:
- A) Using fewer tokens
 - B) Being faster to execute
 - C) Requiring no examples
 - D) Exploring multiple reasoning branches and allowing backtracking
18. The **IRAC** framework (Issue, Rule, Analysis, Conclusion) is specifically designed for:
- A) Medical diagnosis
 - B) Financial reporting
 - C) Software development
 - D) Legal reasoning and analysis
19. The term **Automation Bias** refers to:

- A) Bias in robotic systems
- B) Preference for automated testing
- C) The tendency for humans to over-rely on AI outputs without critical verification
- D) AI systems preferring automated tasks

20. **Agentic AI** differs from traditional chatbots by being:

- A) Unable to use external tools
- B) Text-only without multimodal capabilities
- C) Proactive, multi-step, stateful, and capable of autonomous tool usage
- D) Slower and less accurate

— END OF EXAM —

Optical Form Answer Sheet

Student Name: _____ Student ID: _____

Instructions: Fill in the circle completely for your answer choice. Use pencil only.

Q	A	B	C	D	Q	A	B	C	D
1	☐	☐	☐	☐	11	☐	☐	☐	☐
2	☐	☐	☐	☐	12	☐	☐	☐	☐
3	☐	☐	☐	☐	13	☐	☐	☐	☐
4	☐	☐	☐	☐	14	☐	☐	☐	☐
5	☐	☐	☐	☐	15	☐	☐	☐	☐
6	☐	☐	☐	☐	16	☐	☐	☐	☐
7	☐	☐	☐	☐	17	☐	☐	☐	☐
8	☐	☐	☐	☐	18	☐	☐	☐	☐
9	☐	☐	☐	☐	19	☐	☐	☐	☐
10	☐	☐	☐	☐	20	☐	☐	☐	☐

Instructor Signature: _____ Date: _____

Answer Key (For Instructor Use Only)

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	6	D	11	B	16	C
2	D	7	A	12	A	17	D
3	D	8	C	13	C	18	D
4	A	9	D	14	C	19	C
5	C	10	C	15	C	20	C

Detailed Answer Key with Explanations

1. **D** - A prompt is the input message given to an AI system.
2. **D** - Tokenization breaks text into processable chunks.
3. **D** - Embeddings are numerical vectors capturing semantic meaning.
4. **A** - The Transformer architecture introduced self-attention.
5. **C** - Hallucination is generating plausible but incorrect information.
6. **D** - Output Specification defines format like JSON, tables.
7. **A** - Zero-shot means no examples, just the task.
8. **C** - Few-shot steers style/format with examples.
9. **D** - Conflicting directives are a common anti-pattern.
10. **C** - RAG grounds answers with provided context.
11. **B** - CO-STAR: Context, Objective, Style, Tone, Audience, Response.
12. **A** - Meta-prompting uses prompts to generate/refine prompts.
13. **C** - Field (Topic Domain) controls terminology density.
14. **C** - Brand Voice Fingerprinting extracts measurable attributes.
15. **C** - LLM-as-Judge enables scalable evaluation.
16. **C** - Verbosity Bias favors longer answers.
17. **D** - ToT explores multiple branches with backtracking.
18. **D** - IRAC is for Legal reasoning.
19. **C** - Automation Bias is over-reliance on AI outputs.
20. **C** - Agentic AI is proactive, multi-step, stateful.

Topic Coverage by Question

- **Foundations** (Q1–Q5): Basic concepts of prompts, tokenization, embeddings, transformers, and hallucination
- **Prompt Design Principles** (Q6–Q10): Output specification, zero-shot, few-shot, anti-patterns, and RAG
- **Patterns and Frameworks** (Q11–Q12): CO-STAR framework and meta-prompting
- **Style and Format Control** (Q13–Q14): Linguistic register and brand voice fingerprinting
- **Evaluation Methods** (Q15–Q16): LLM-as-Judge and verbosity bias
- **Complex Problem Solving** (Q17): Tree of Thoughts
- **Domain-Specific Applications** (Q18): IRAC framework for legal reasoning
- **Ethics and Responsible AI** (Q19–Q20): Automation bias and agentic AI

Grading Scale

Points	Grade
90–100	AA
85–89	BA
80–84	BB
75–79	CB
70–74	CC
65–69	DC
60–64	DD
50–59	FD
0–49	FF